

Chi Ho YUEN

CONTACT INFORMATION	Dept. of Applied Mathematics, Office 341 Science Bld. 1, NYCU 1001 Daxue Rd., East District Hsinchu 30009, Taiwan	chyuen@math.nctu.edu.tw https://yuenchiho.github.io/
EMPLOYMENT	2024 – current 11/23 – 01/24 2021 – 2023 2019 – 2021 2018 – 2019	National Yang Ming Chiao Tung University Assistant Professor (Tenure-track) University of Copenhagen Postdoctoral Researcher University of Oslo Postdoctoral Researcher Brown University Croucher Postdoctoral Fellow University of Bern Postdoctoral Researcher
EDUCATION	2012 – 2018 2008 – 2012	Georgia Institute of Technology Ph.D. in Algorithms, Combinatorics, and Optimization Advisor: Matthew Baker Thesis: Geometric Bijections of Graphs and Regular Matroids Minor: Algorithms with Randomness The Chinese University of Hong Kong B.Sc. in Mathematics (First Class Honor) B.Eng. in Information Engineering (First Class Honor)
PUBLICATIONS	White’s Conjecture for Paving Matroids. With Yu-Chuan Yu. Submitted. Preprint available at arXiv:2510.04163. Period Matrices and Homological Quasi-trees on Discrete Riemann Surfaces. With Wai Yeung Lam and On-Hei Solomon Lo. Accepted by <i>Journal für die Reine und Angewandte Mathematik (Crelle’s Journal)</i> . Filtrations of Tope Spaces of Oriented Matroids. With Kris Shaw. Accepted by <i>Transactions of the American Mathematical Society</i> . A Consistent Sandpile Torsor Algorithm for Regular Matroids. With Changxin Ding, Alex McDonough, and Lilla Tóthmérész. <i>European Journal of Combinatorics</i> 130 , Paper No. 104218, 2025. Oriented Matroids from Triangulations of Products of Simplices. With Marcel Celaya and Georg Loho. <i>Selecta Mathematica, New Series</i> 30 , Art. 47, 32pp, 2024. The Amoeba Dimension of a Linear Space. With Jan Draisma, Sarah Eggleston, Rudi Pendavingh, and Johannes Rau. <i>Proceedings of the American Mathematical Society</i> 152 , 2385–2401, 2024. Critical Groups of Adinkras up to 2-Rank of Cayley Graphs. <i>Electronic Journal of Combinatorics</i> 31 , Paper No. 1.38, 9pp, 2024. Sandpile Groups of Cayley Graphs of $\mathbb{F}_2^r$. With Jiyang Gao, Jared Marx-Kuo, and Vaughan McDonald. <i>Communications in Algebra</i> 52 , 4459–4479, 2024.	

Eigenvalues and Critical Groups of Adinkras.
 With Kevin Iga, Caroline Klivans, and Jordan Kostiuk.
Advances in Applied Mathematics **143**, Paper No. 102450, 45pp, 2023.

Patchworking Oriented Matroids.
 With Marcel Celaya and Georg Loho.
Journal of the London Mathematical Society **106**, 3545–3576, 2022.

Critical Groups of Strongly Regular Graphs and Their Generalizations.
 With Kenneth Hung.
Innovations in Incidence Geometry: Algebraic, Topological and Combinatorial **19**, 95–109, 2022.

The Dimension of an Amoeba.
 With Jan Draisma and Johannes Rau.
Bulletin of the London Mathematical Society **52**, 16–23, 2020.

Extension-lifting Bijections Bijections for Oriented Matroids.
 With Spencer Backman and Francisco Santos.
Discrete & Computational Geometry **75**, 969–994, 2026.

On the Number of Circuit-cocircuit Reversal Classes of an Oriented Matroid.
 With Emeric Gioan.
Discrete Mathematics **342**, 1056–1059, 2019.

Geometric Bijections for Regular Matroids, Zonotopes, and Ehrhart Theory.
 With Spencer Backman and Matthew Baker.
Forum of Mathematics, Sigma **7**, e45, 37pp, 2019.

Geometric Bijections Between Spanning Trees and Break Divisors.
Journal of Combinatorial Theory, Series A **152**, 159–189, 2017.

SELECTED TALKS

Filtrations of Tope Spaces of Oriented Matroids

NCTS Tropical Geometry in Taiwan. (August 2025)
 IBS Workshop on (Mostly) Matroids. (August 2024)
 MIT-Harvard Combinatorics Seminar. (September 2023)
 QMUL Workshop on Matroids and Tropical Combinatorics. (January 2023)
 KTH Combinatorics Seminar. (November 2022)

The Critical Groups of Adinkras and Similar Objects

ILAS 2025. (June 2025)
 Brandeis Combinatorics Seminar. (November 2021)

Oriented Matroids from Triangulations of Products of Simplices

BIRS Workshop on Algebraic Aspects of Matroid Theory. (March 2023)
 CAS Workshop on Real Structures in Geometry. (August 2021)

The Dimension of an Amoeba

Scandinavian Gathering Around Remarkable Discrete Mathematics. (May 2019)
 Georgia Tech Algebra Seminar. (January 2019)

Geometric Bijections of Graphs and Regular Matroids

CIRM Workshop on Combinatorial Geometries. (September 2018)
 Institut Mittag-Leffler Seminar. (February 2018)
 MIT Combinatorics Seminar. (October 2017)
 Brown University Discrete Math Seminar. (May 2017)
 BIRS-CMO Workshop on Sandpile Groups. (November 2015)
 Cornell Combinatorics and Discrete Geometry Seminar. (April 2015)

SELECTED TEACHING EXPERIENCES	Lecturer at NYCU	
	Spring 26	Calculus A (II)
	Spring 26	Introduction to Combinatorics
	Fall 25	Calculus A (I)
	Spring 25	Calculus A (II)
	Spring 25	Introduction to Combinatorics
	Fall 24	Calculus A (I)
	Spring 24	Calculus A (II)
	Spring 24	Discrete Mathematics
	Lecturer at University of Copenhagen	
	Block 2 23-24	Invitation to Combinatorics
	Lead Instructor at Georgia Tech	
	Summer 18	Introduction to Discrete Mathematics
GRADUATE STUDENTS	Fall 16	Grad Level Graph Theory (Co-lecturing with Robin Thomas)
	Assistant at University of Bern	
	Math. I für Naturwiss.; Algorithms in Algebra; Tensors, Geometry and Applications	
	Teaching Assistant at Georgia Tech	
	Calculus II & III; Linear Algebra with Abstract Vector Spaces; Intro. to Discrete Math.	
	At NYCU	
	2024 –	Yu-Chuan Yu (master student)
	OTHER MENTORING At NYCU	
	Summer 24	Summer Research Internship of P.H. Chen and W.S. Wu
	At University of Oslo	
	Spring 23	Undergraduate Project (MAT2000) of Trond Skaret Johansen
	At Brown University	
	2020–2021	Senior Honor Thesis of Nathan Zelesko (Co-advising with Melody Chan)
RESEARCH GRANT	2025 – 2028	NSTC Grant MOST 114-2115-M-A49-015-MY3
	2024 – 2025	NSTC Grant MOST 113-2115-M-A49-004-MY2
SELECTED HONORS, AWARDS AND SCHOLARSHIPS	2019 – 2021	Croucher Fellowship for Postdoctoral Research
	2018 – 2019	Georgia Tech School of Mathematics Best PhD Thesis Award
	Spring 2018	Institut Mittag-Leffler Fellowship
	Summer 2011	Caltech – Hong Kong Undergraduate Research Fellowship
ORGANIZED EVENTS	Co-organizer of the first <i>MaP-Trop</i> seminar series at KTH and UiO (Jan 2024).	
	Co-organizer of the minisymposium <i>Chip-firing and Tropical Curves</i> at SIAMAG 19.	
REFeree DUTIES	PNAS, Forum of Mathematics, Sigma, Journal of the LMS, Research in the Mathematical Sciences, SIAM Journal on Applied Algebra and Geometry, Combinatorial Theory, Combinatorica, Algebraic Combinatorics, Advances in Combinatorics, SIAM Journal on Discrete Mathematics, European Journal of Combinatorics, Electronic Journal of Combinatorics, Annals of Combinatorics, Discrete Mathematics, Discrete Applied Mathematics, FPSAC International Conference, and American Mathematical Monthly.	
OTHER SERVICES	Coordinator of the 57th International Mathematical Olympiad.	